

A Gentle Introduction to Diffusion Models



EE367/CS448L: Computational Imaging

stanford.edu/class/ee367/

Lecture 12

Slides based on:

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Stanford University

“Creating noise from data is easy; creating data from noise is generative modeling.”

–Song et al., ICLR 2021



Disclaimer

- Extremely active area – cannot possibly cover everything in 1 introductory lecture!
- Focus on fundamentals of diffusion models to get you started

Overview

- High-level introduction
- Score-based generative modeling / diffusion models
- Diffusion model architectures & attention mechanisms

Generate 100 images



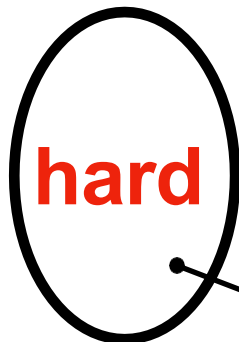
Generate 100 images



Generate 100 images of **raspberries**

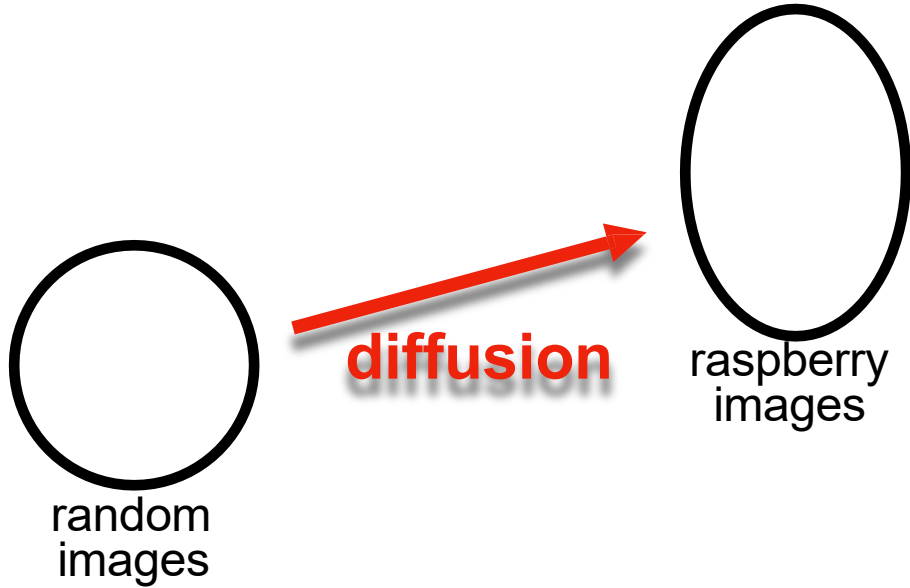


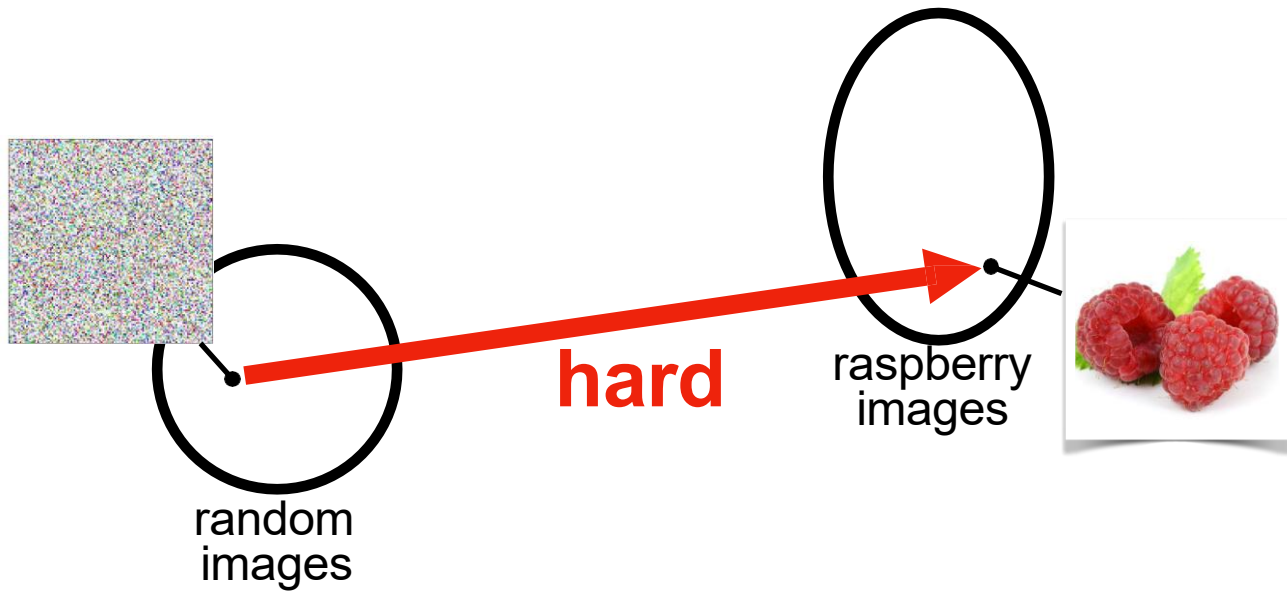
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images

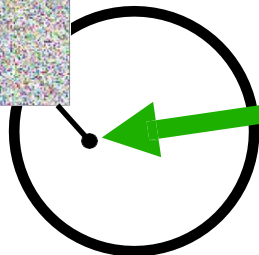


raspberry
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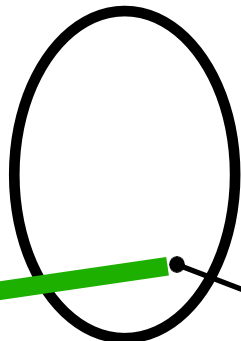






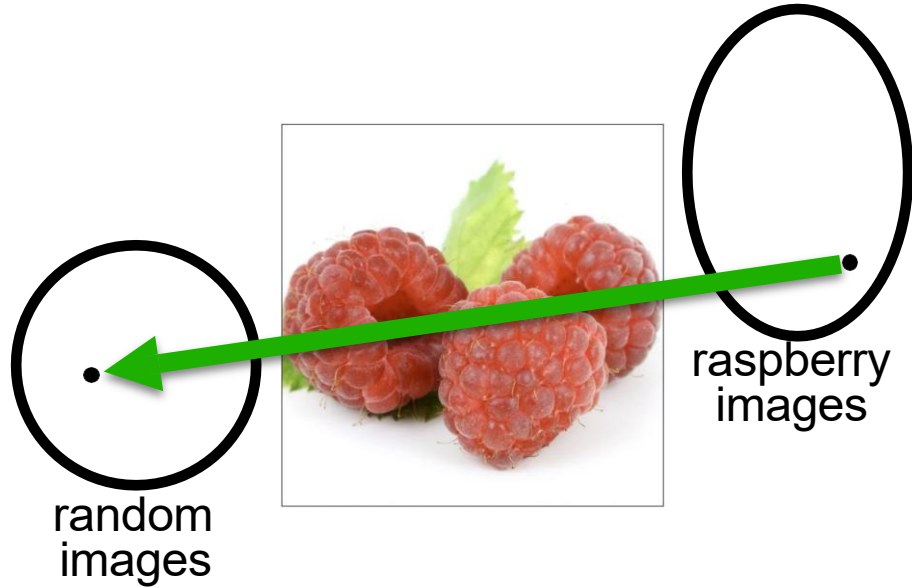
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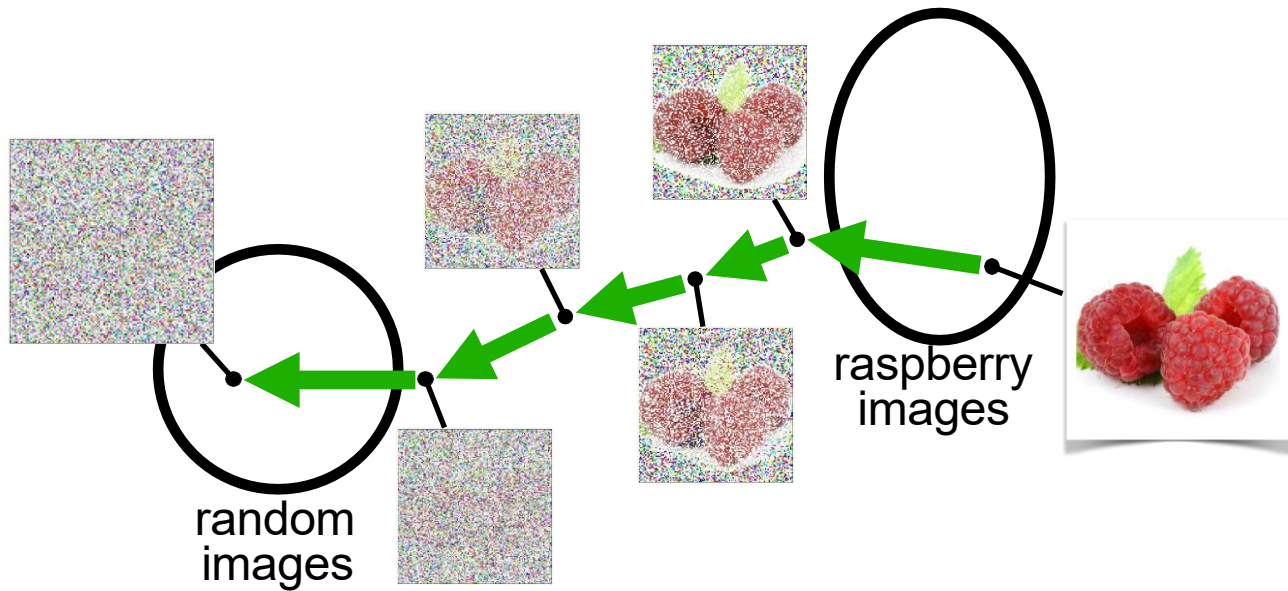
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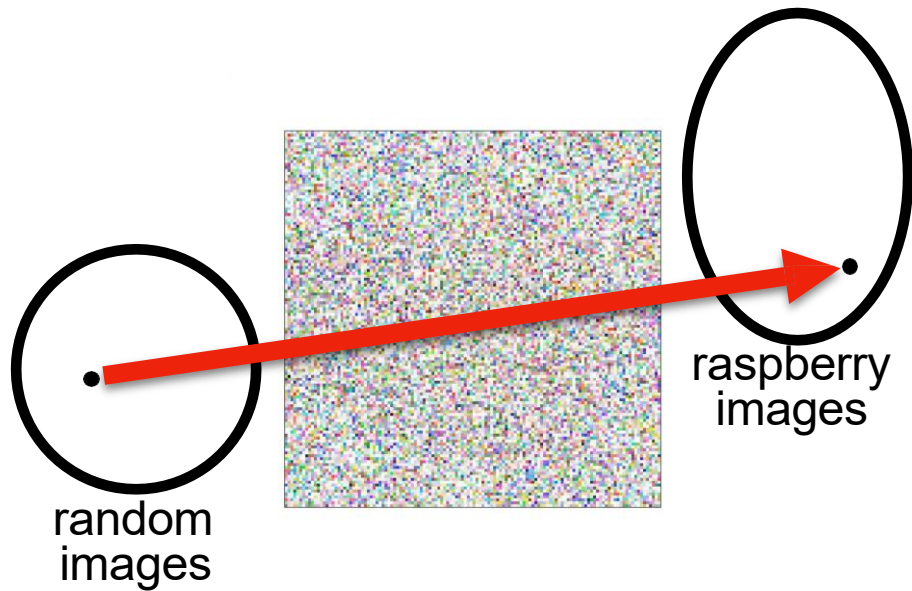


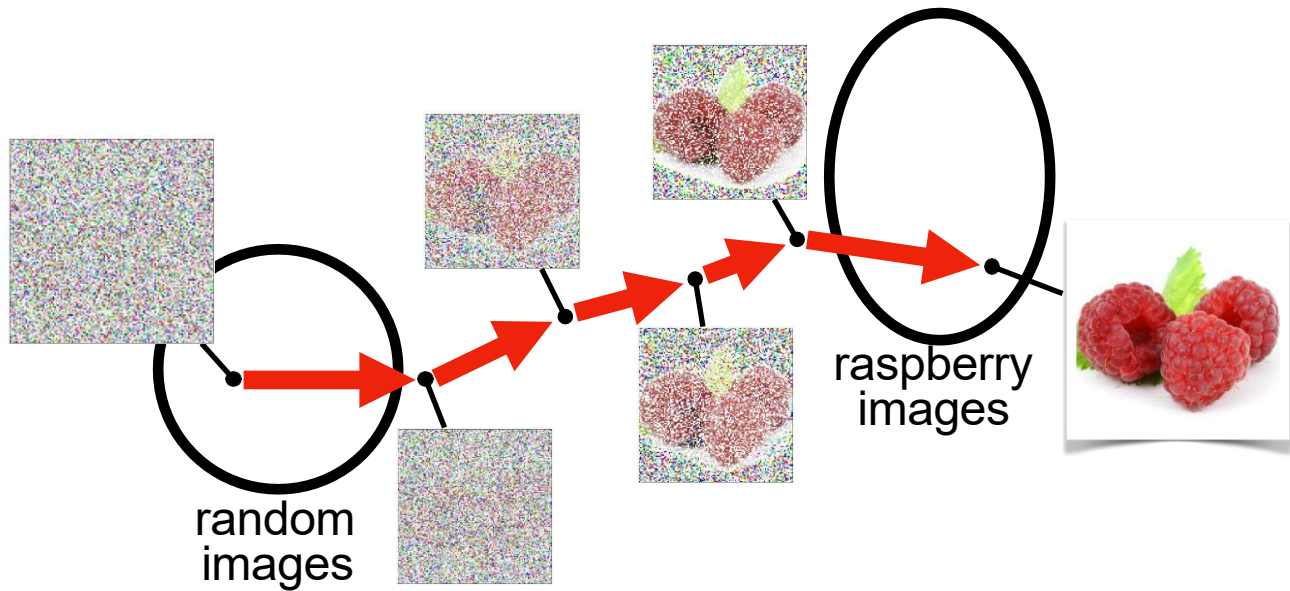
raspberry
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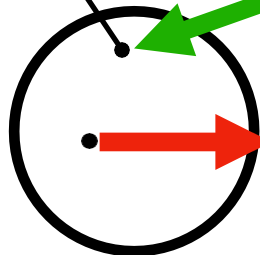




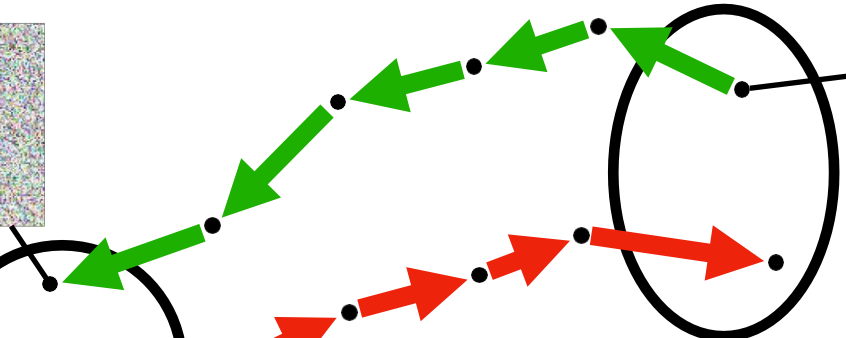






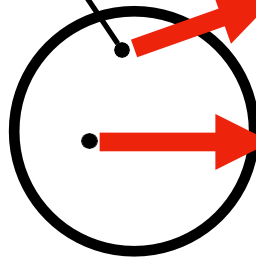


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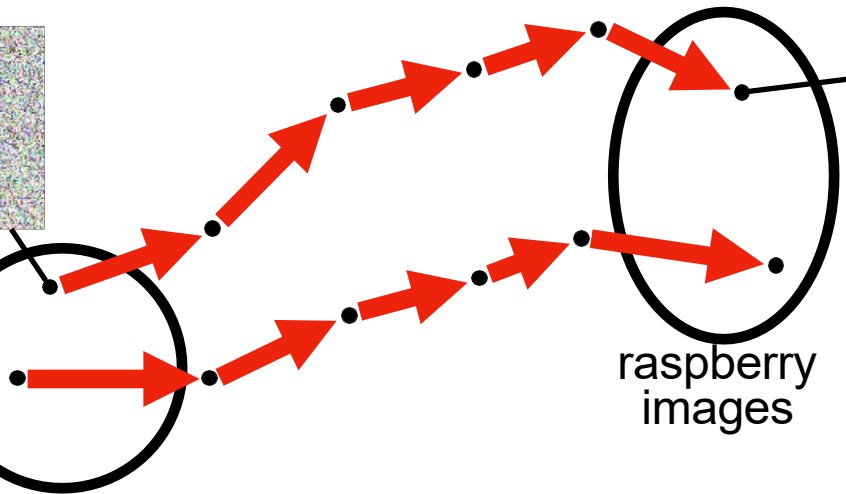


raspberry
images



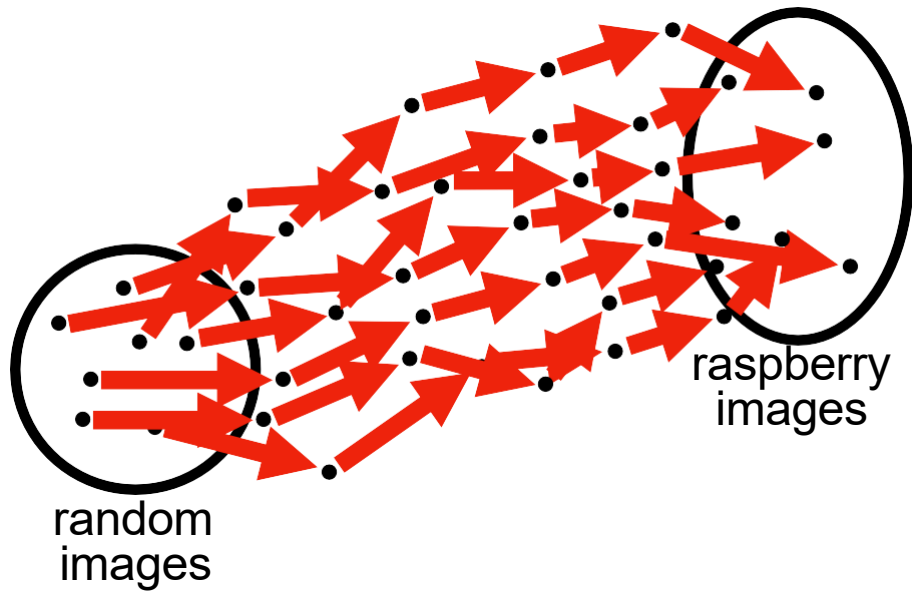


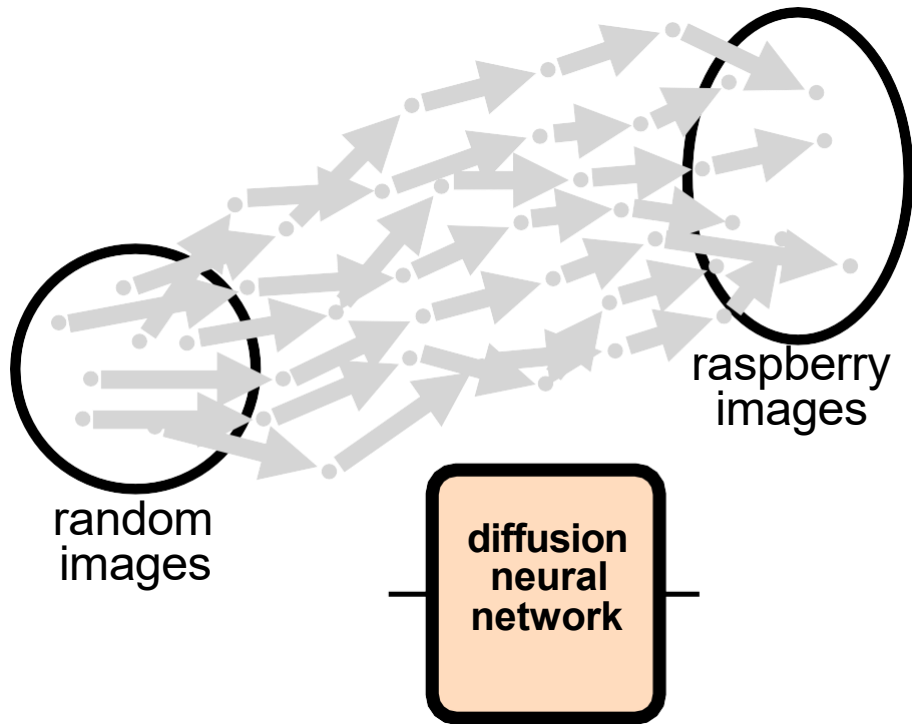
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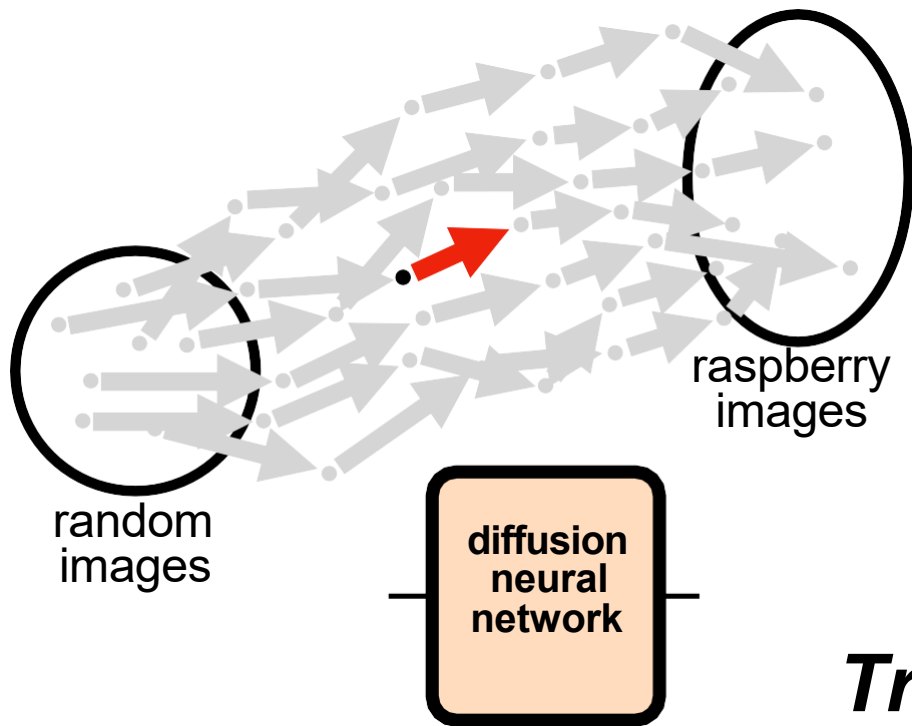


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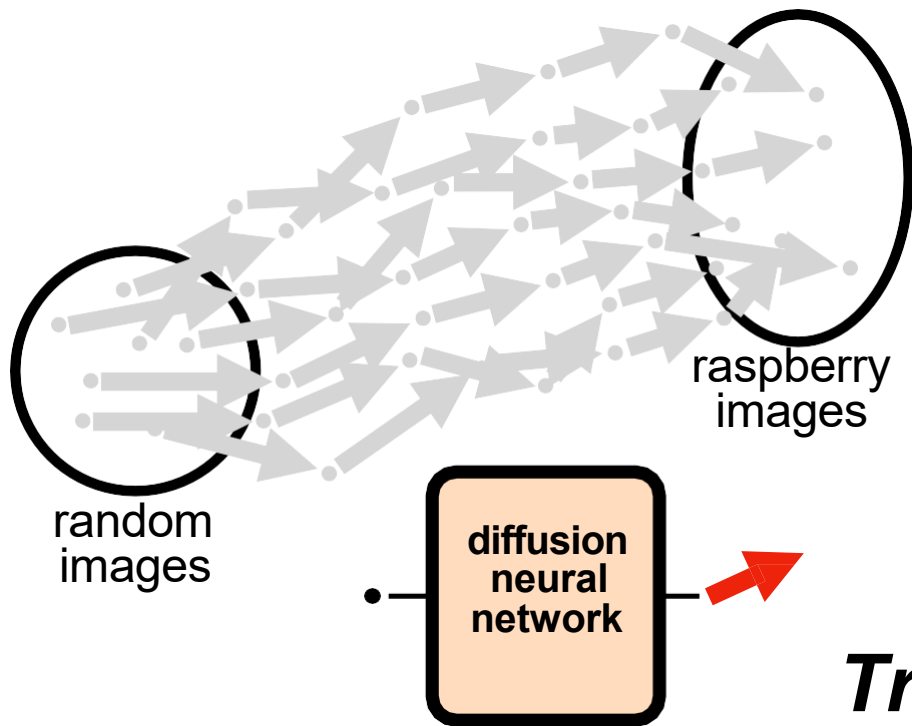


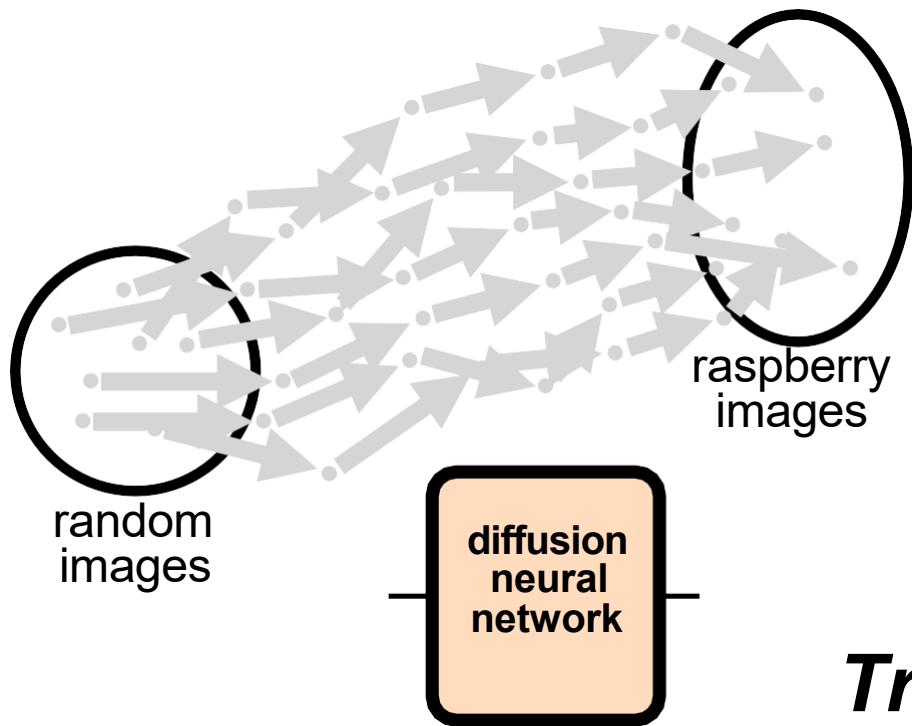




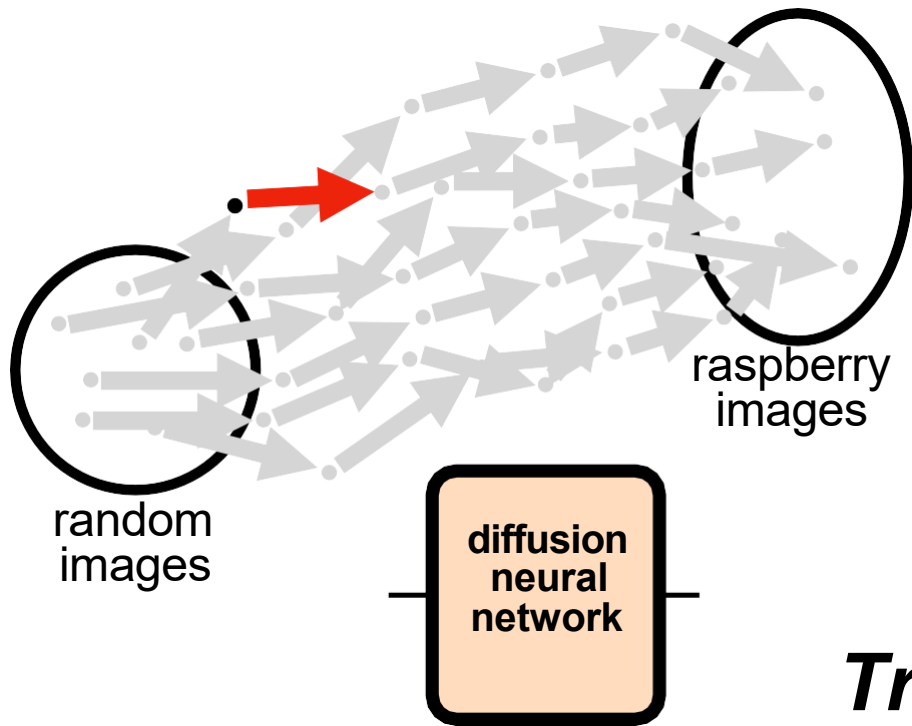


Training

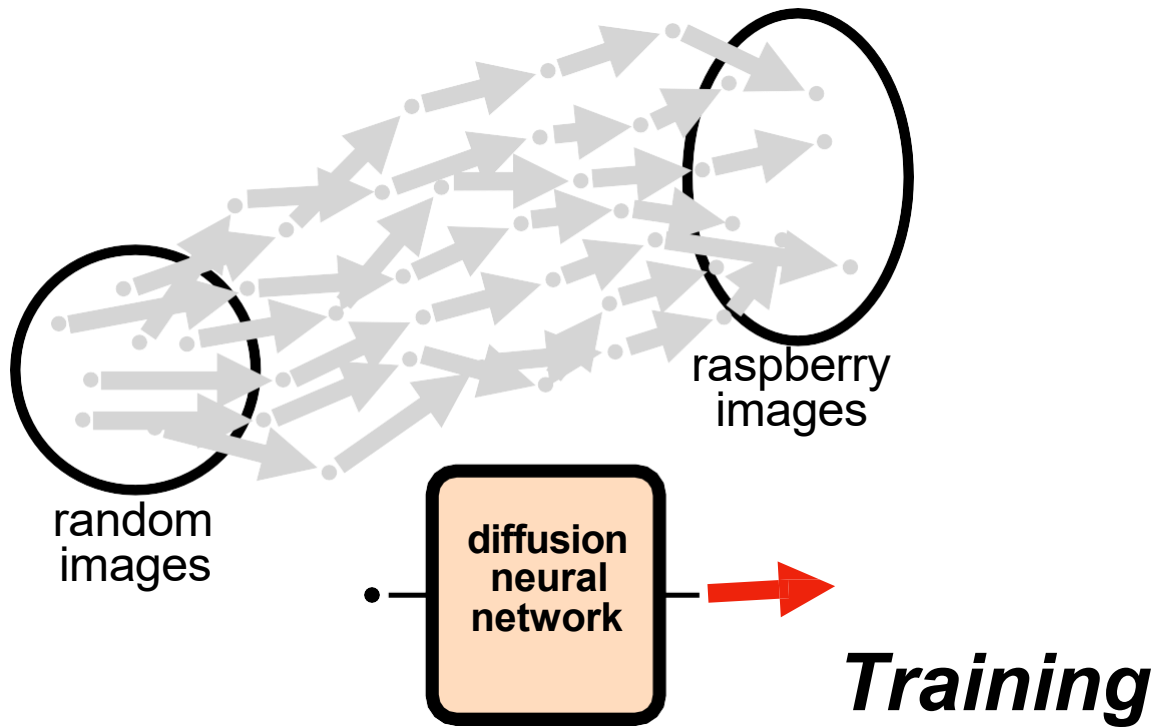


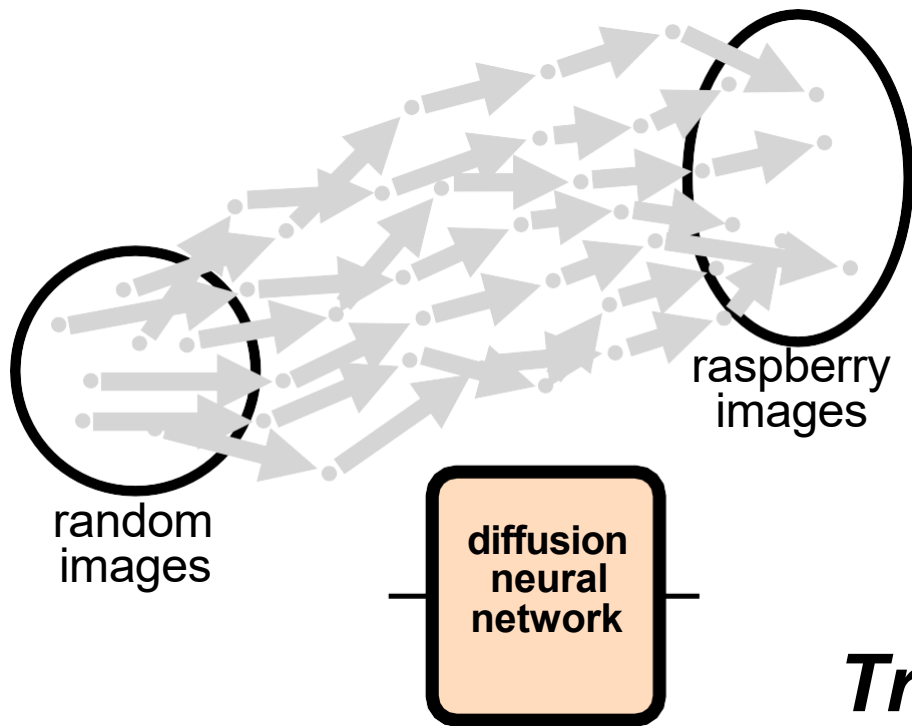


Training

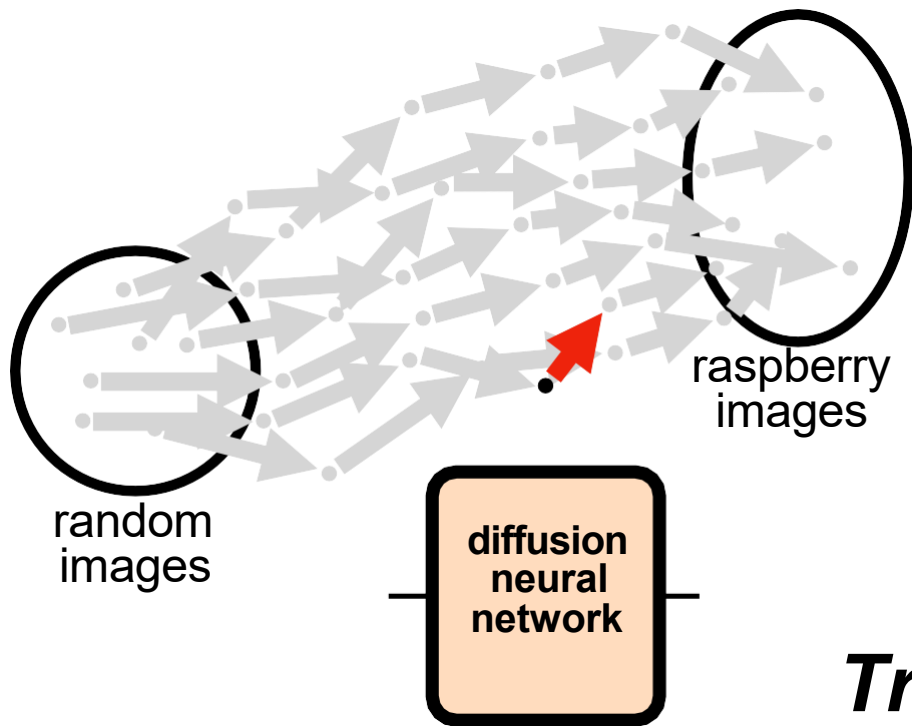


Training

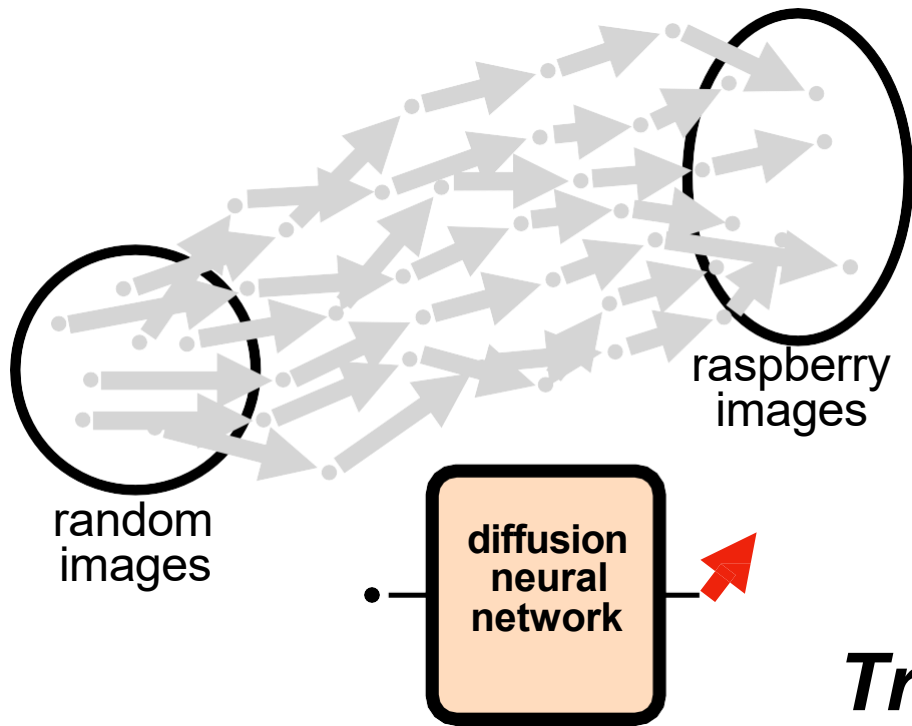




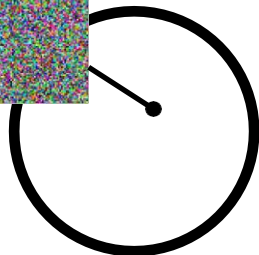
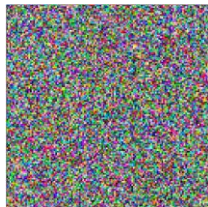
Training



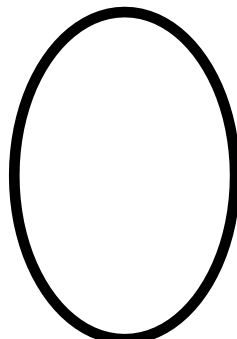
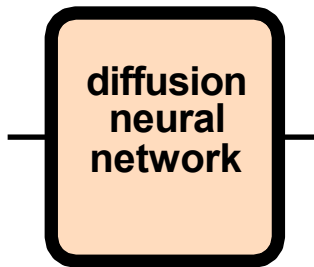
Training



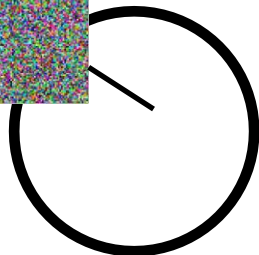
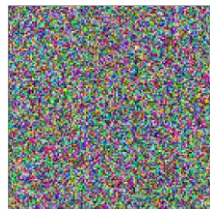
Training



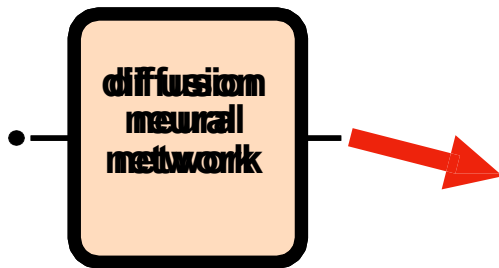
random
images



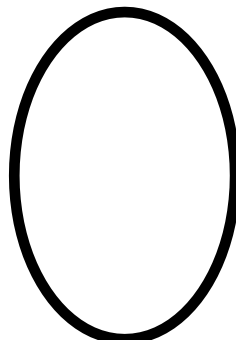
raspberry
images



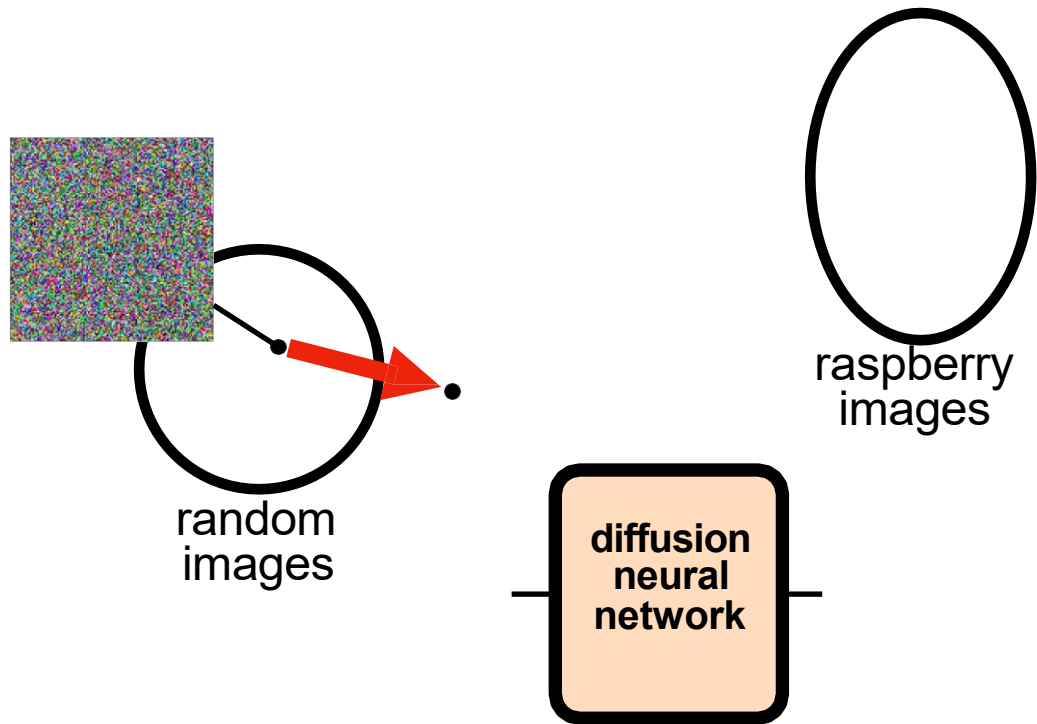
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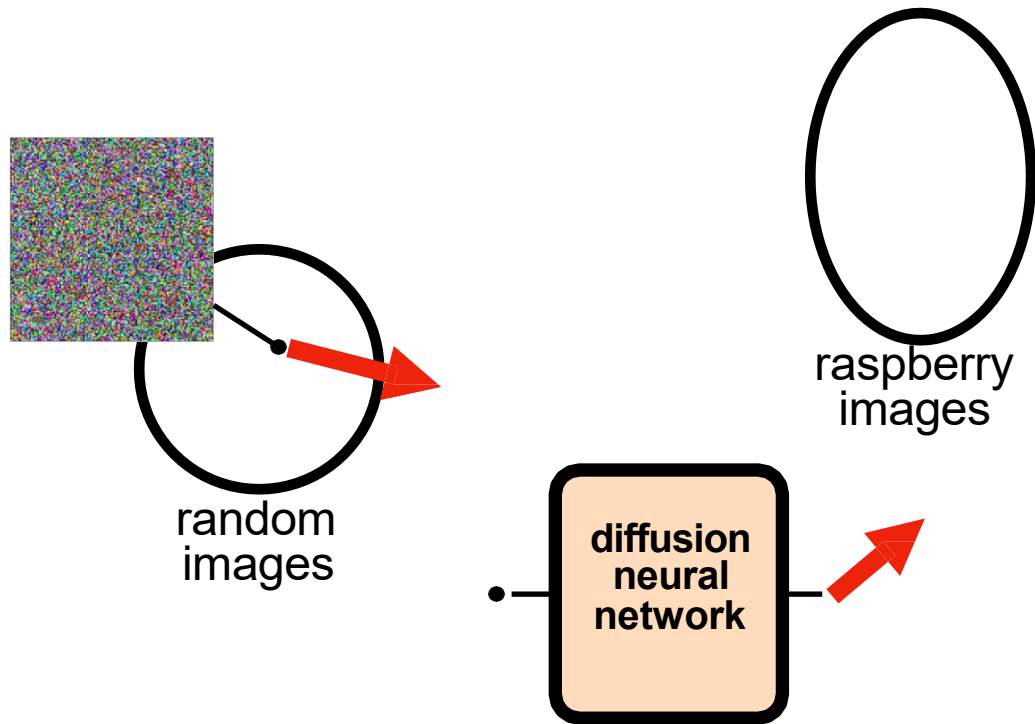


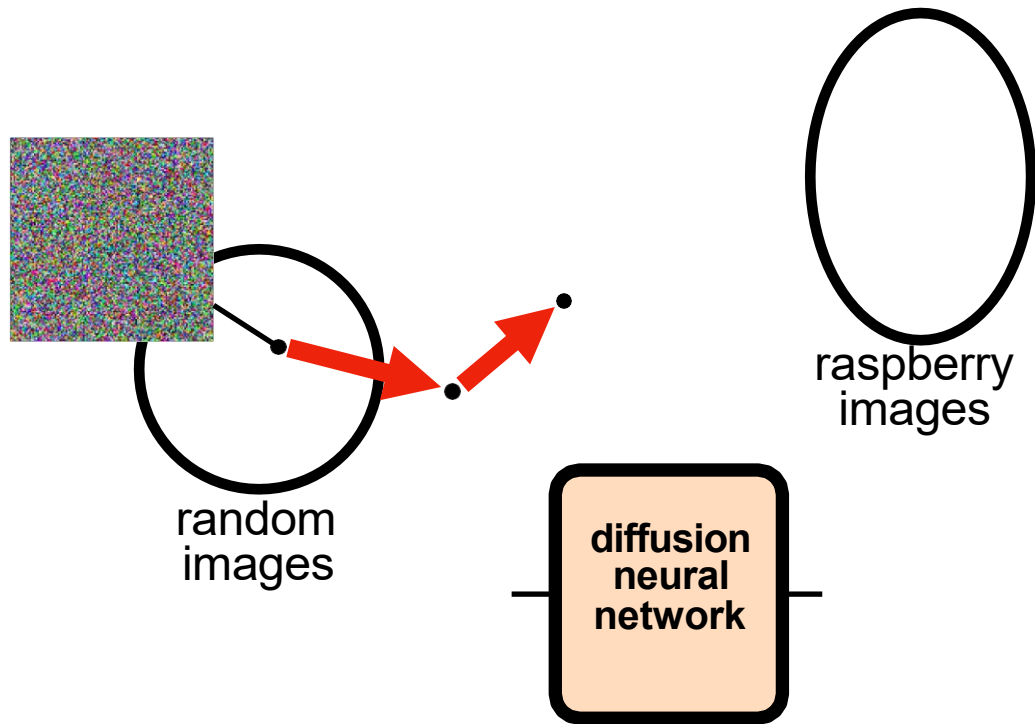
diffusion
neural
network

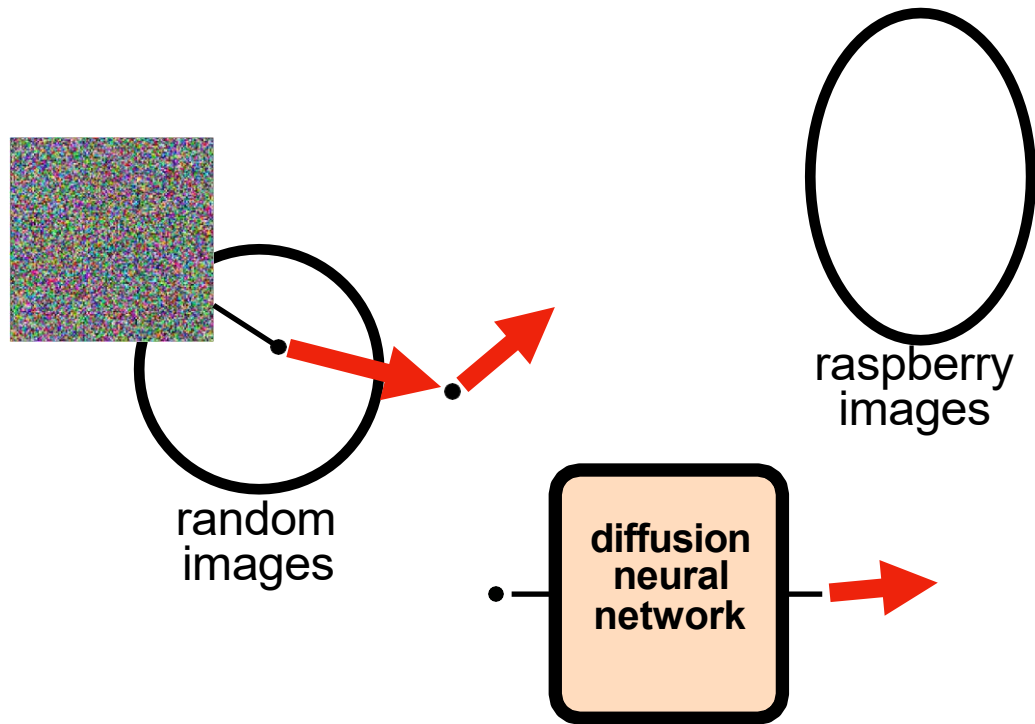


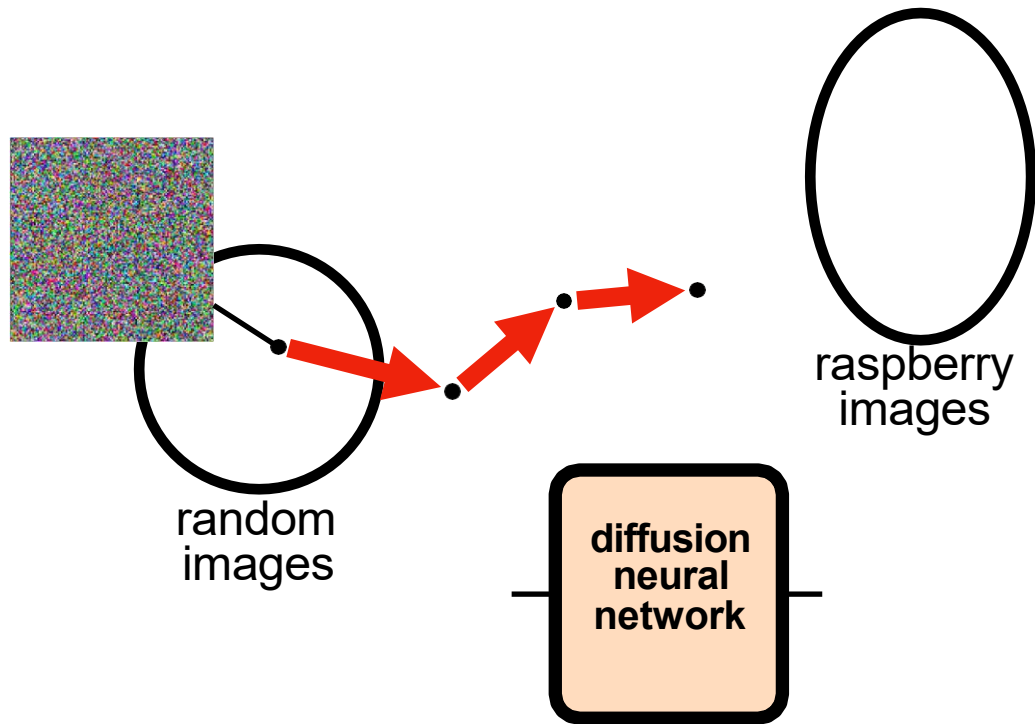
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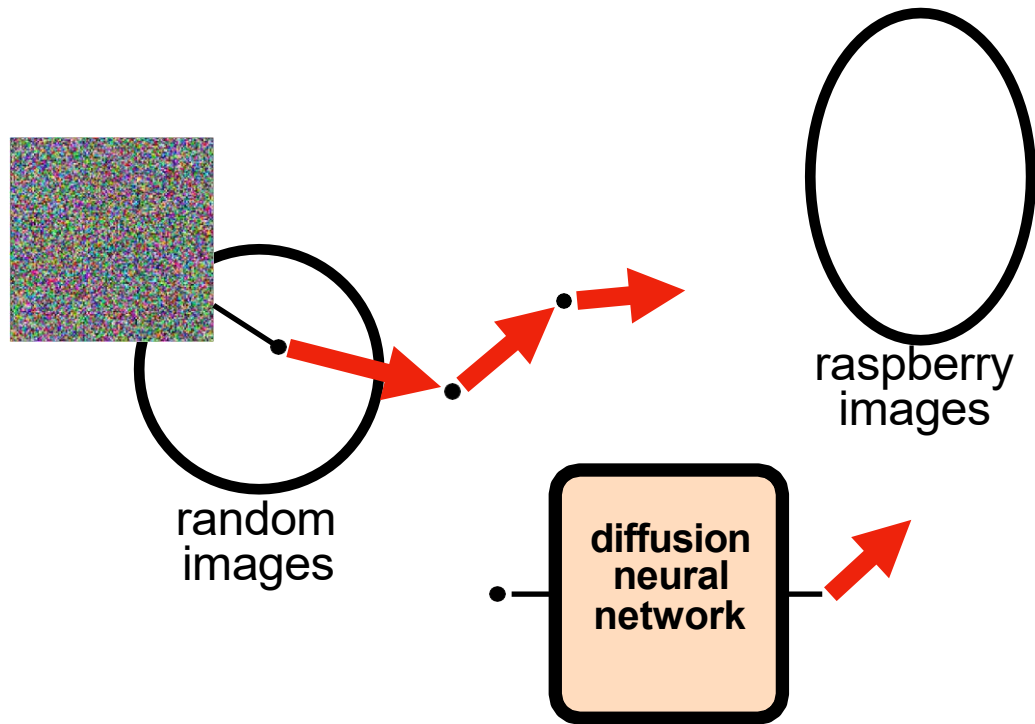


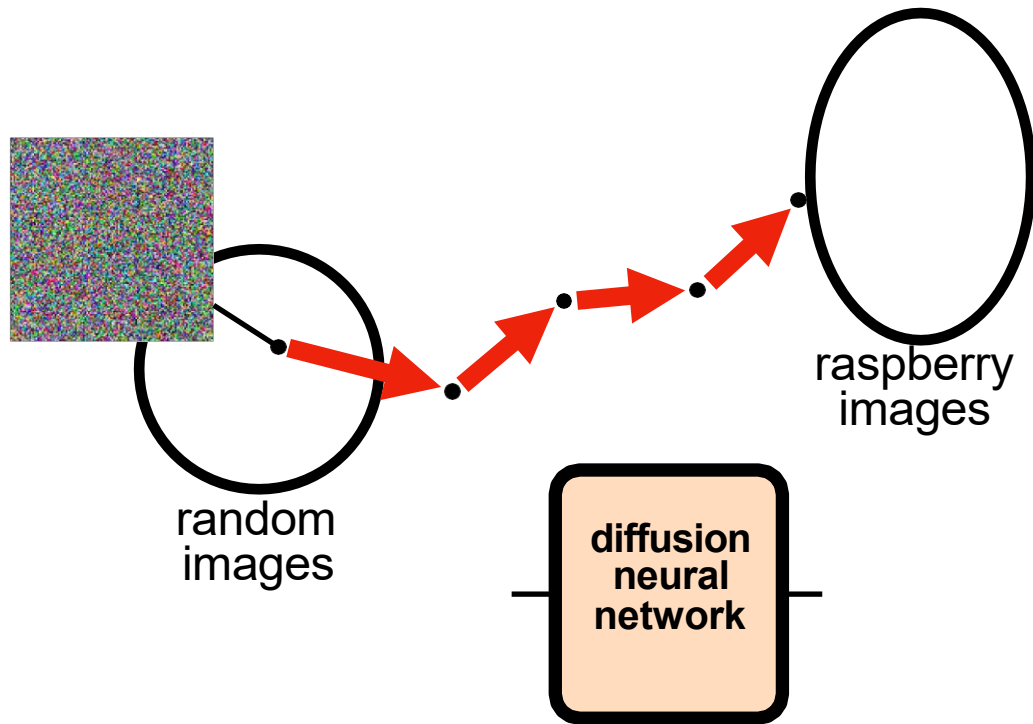


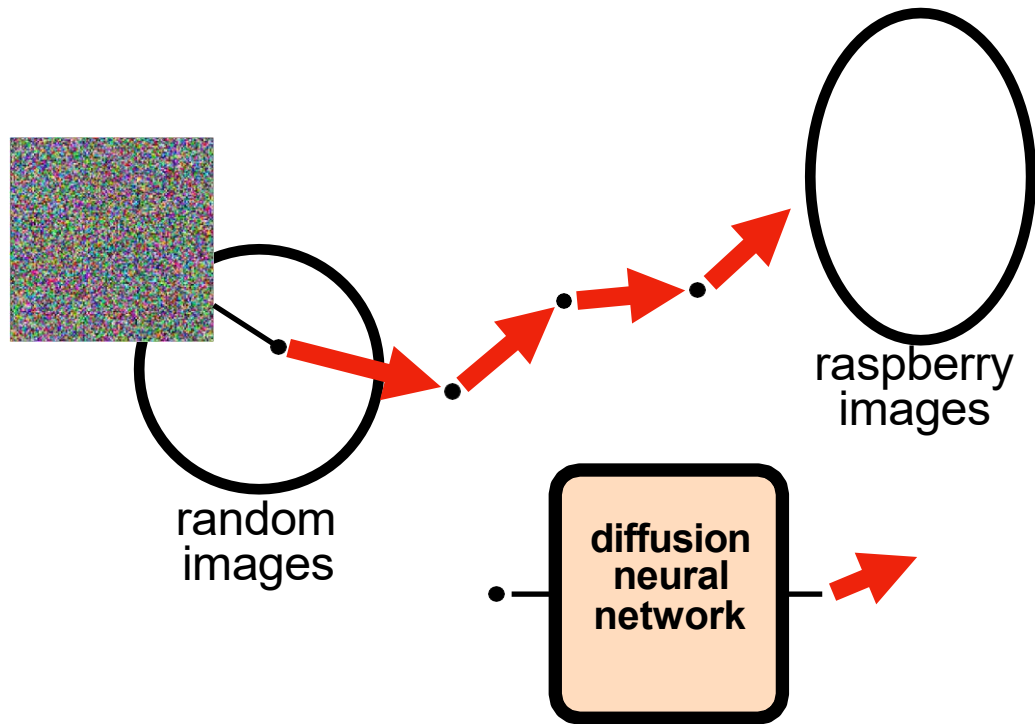


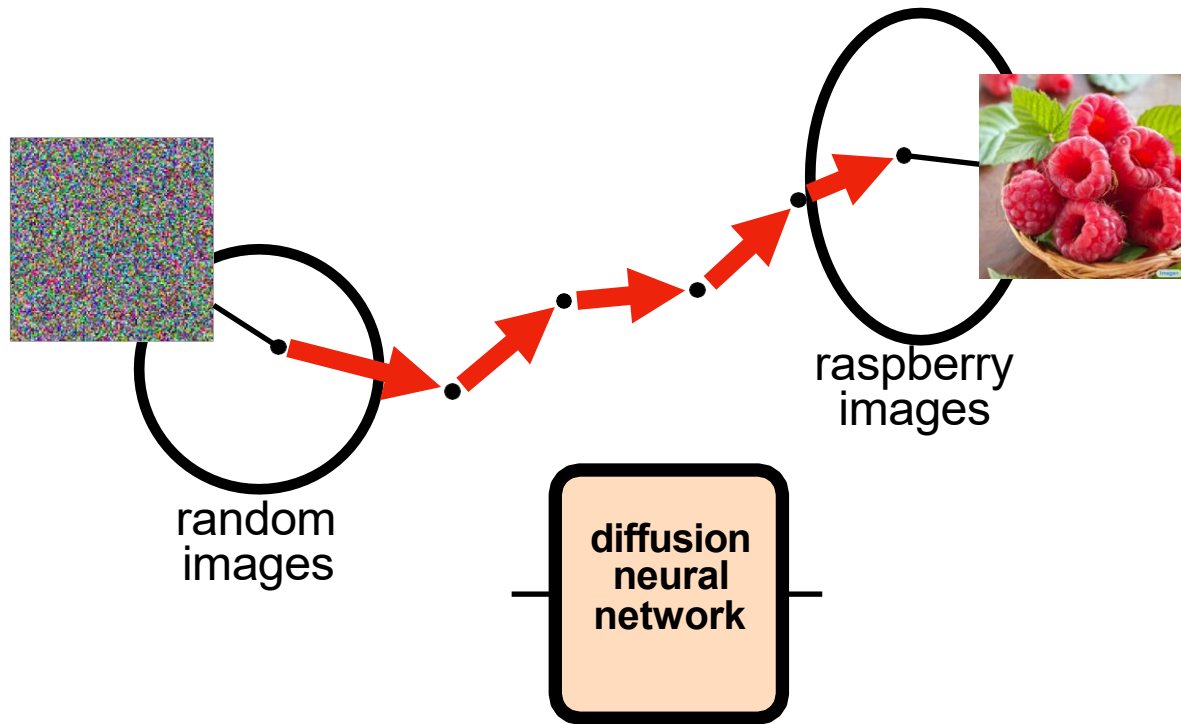


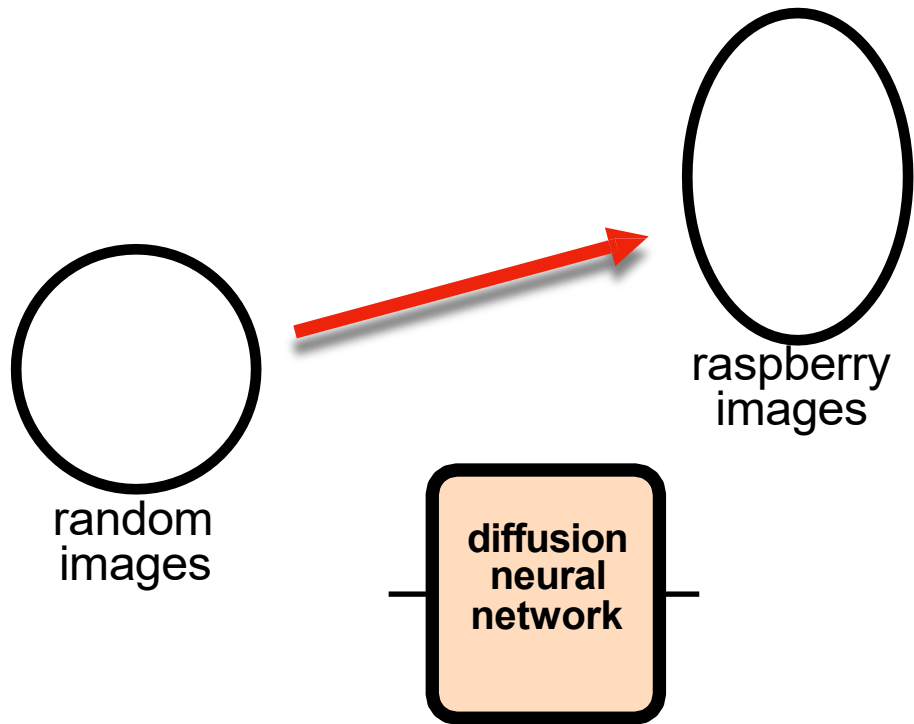


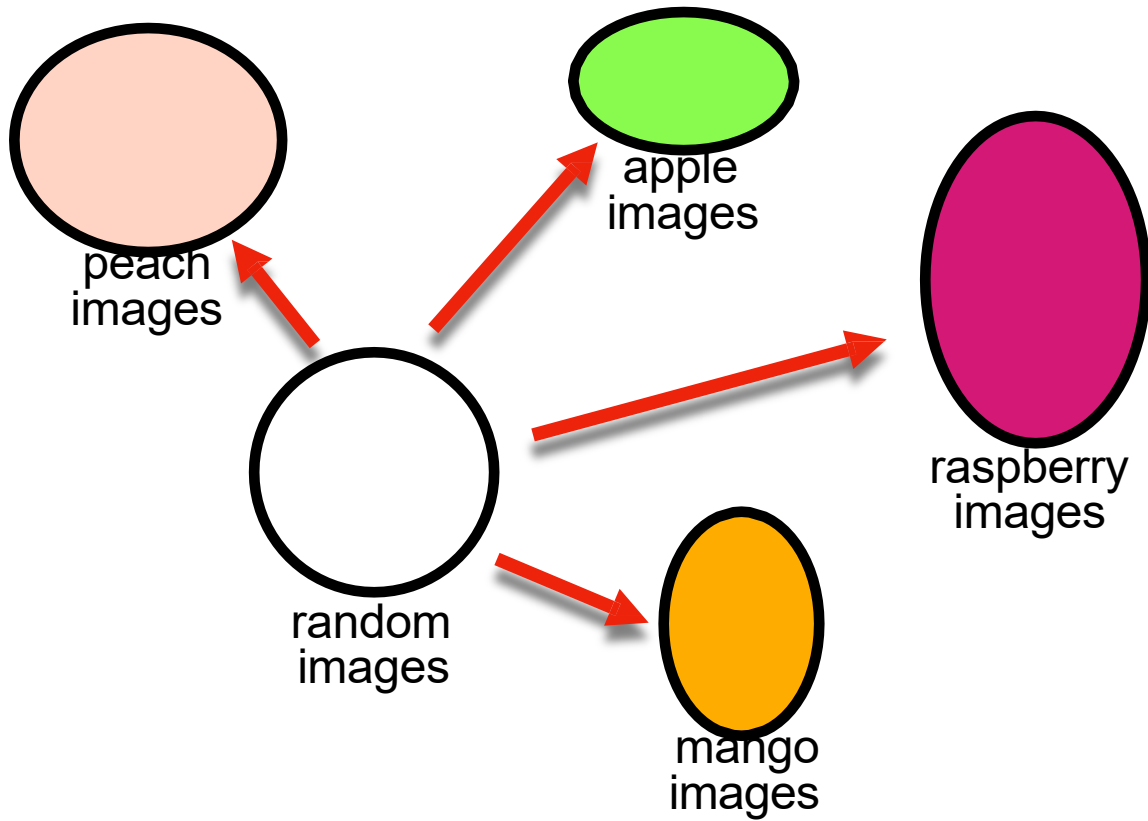


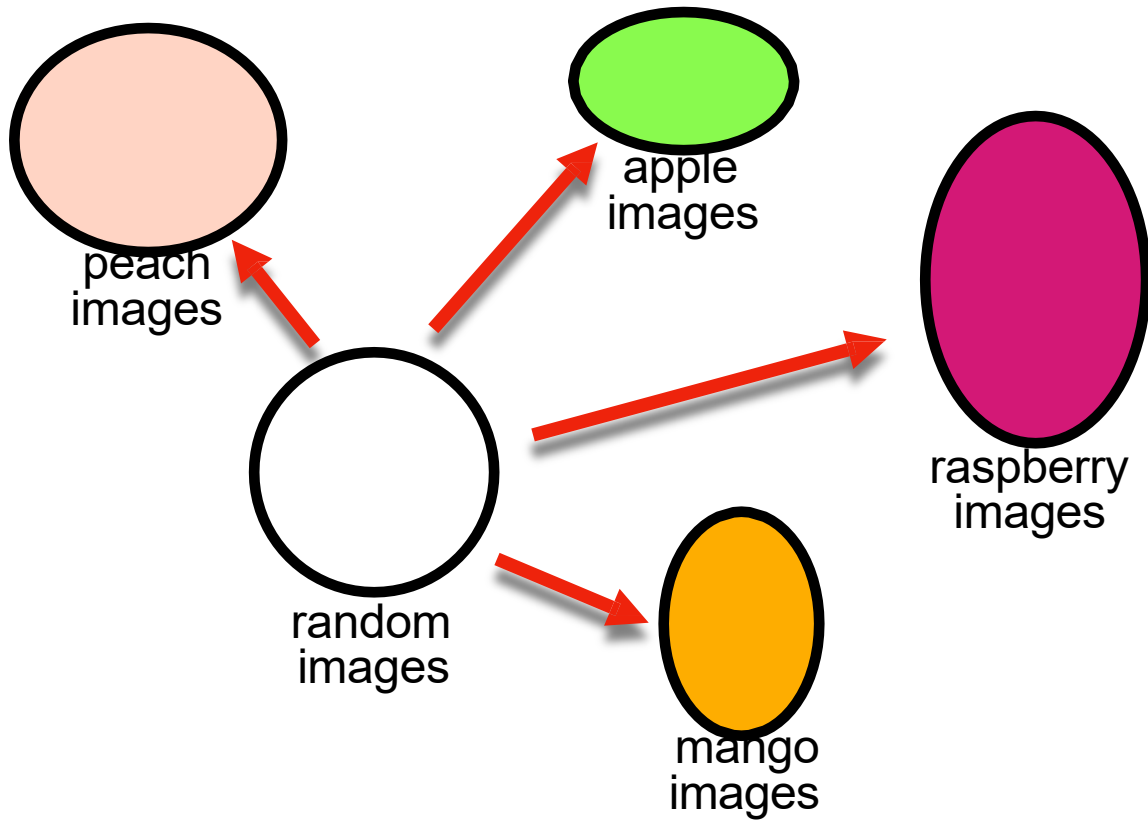


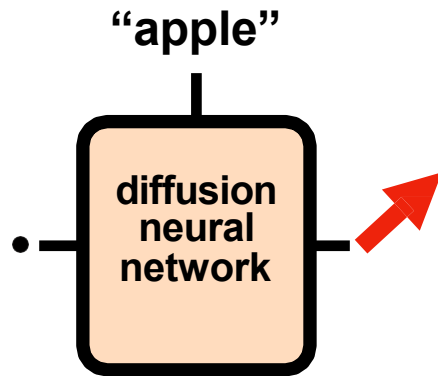
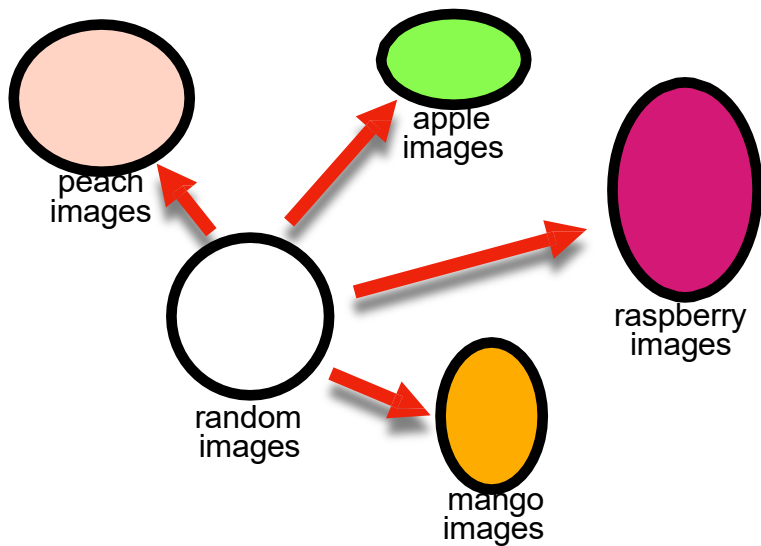


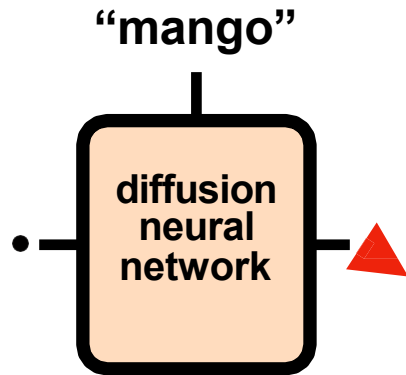
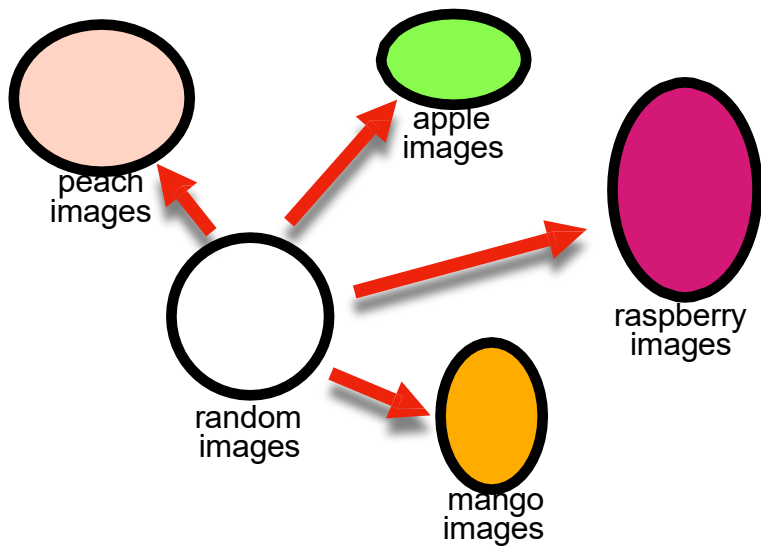




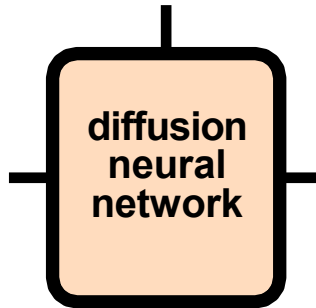


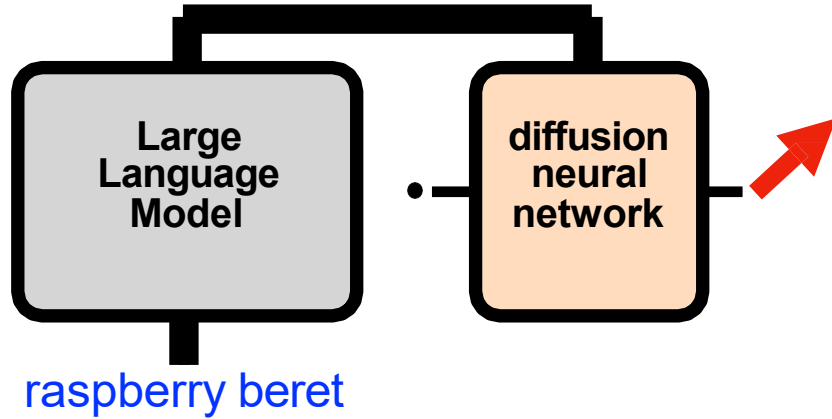






raspberry beret







raspberry beret



raspberry beret

beret of raspberries



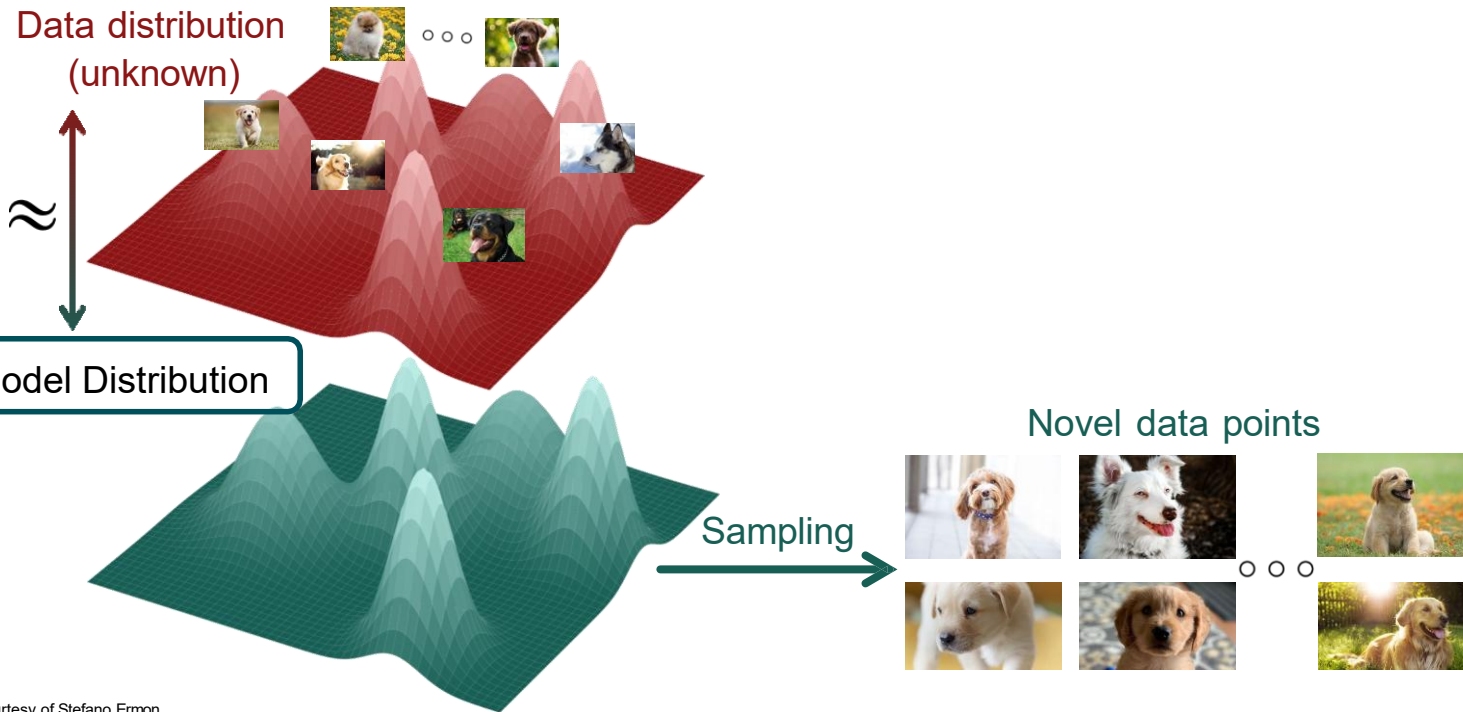
beret of raspberries



beret of raspberries

Score-based Generative Modeling

Estimating the probability distribution of data



Estimating the probability distribution of data

Data distribution
(unknown)



Data distribution is extremely complex for high dimensional data.

Model distribution



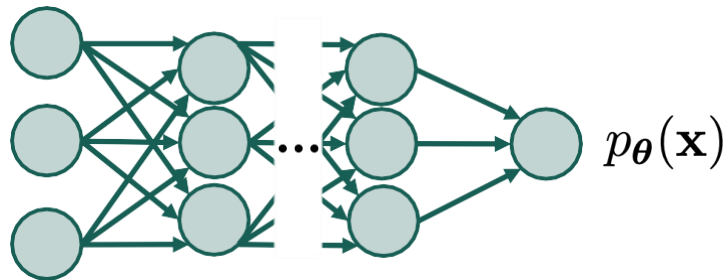
How to build a complex model to fit the data distribution?

Estimating the probability distribution of data

Data distribution
(unknown)

Model distribution

$\mathbf{x}$



Deep Neural Network



Deep Generative Models

Tweedie's Formula

- Famous result in statistics!
- For an additive Gaussian corruption $\underline{x_t} = \underline{x_0} + \underline{\sigma_t z}, z \sim \mathcal{N}(0, I)$


noisy image clean image noise ϵ_t

- Tweedie: $\mathbb{E}[x_0|x_t] = x_t + \sigma_t^2 \nabla_{x_t} \log p_t(x_t)$

- Thus, score function is $\nabla_{x_t} \log p_t(x_t) = \frac{\mathbb{E}[x_0|x_t] - x_t}{\sigma_t^2}$

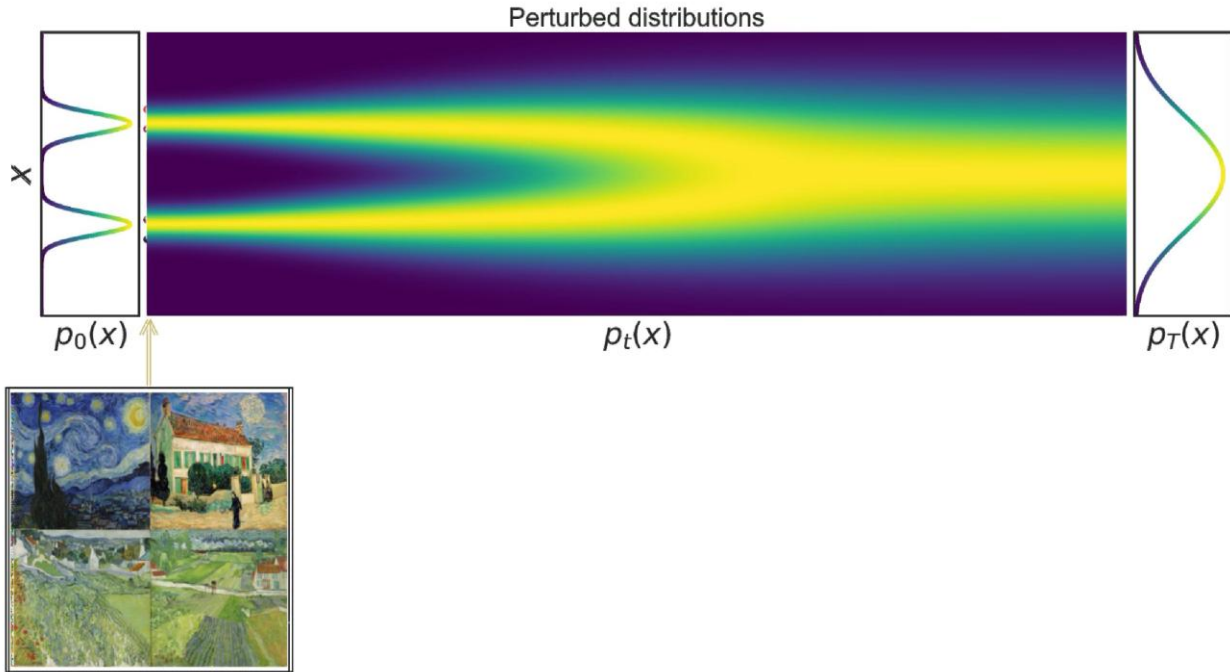
Denoising Score Matching

$$\nabla_{x_t} \log p_t(x_t) = \frac{\mathbb{E}[x_0|x_t] - x_t}{\sigma_t^2} = \underset{\substack{\uparrow \\ \text{score estimation} \\ \text{network}}}{s_\phi(x_t; \sigma_t)} \approx \underset{\substack{\uparrow \\ \text{denoising} \\ \text{network}}}{\frac{D_\theta(x_t; \sigma_t) - x_t}{\sigma_t^2}} = - \underset{\substack{\nwarrow \\ \text{noise estimation} \\ \text{network}}}{\frac{\epsilon_\phi(x_t; \sigma_t)}{\sigma_t^2}}$$

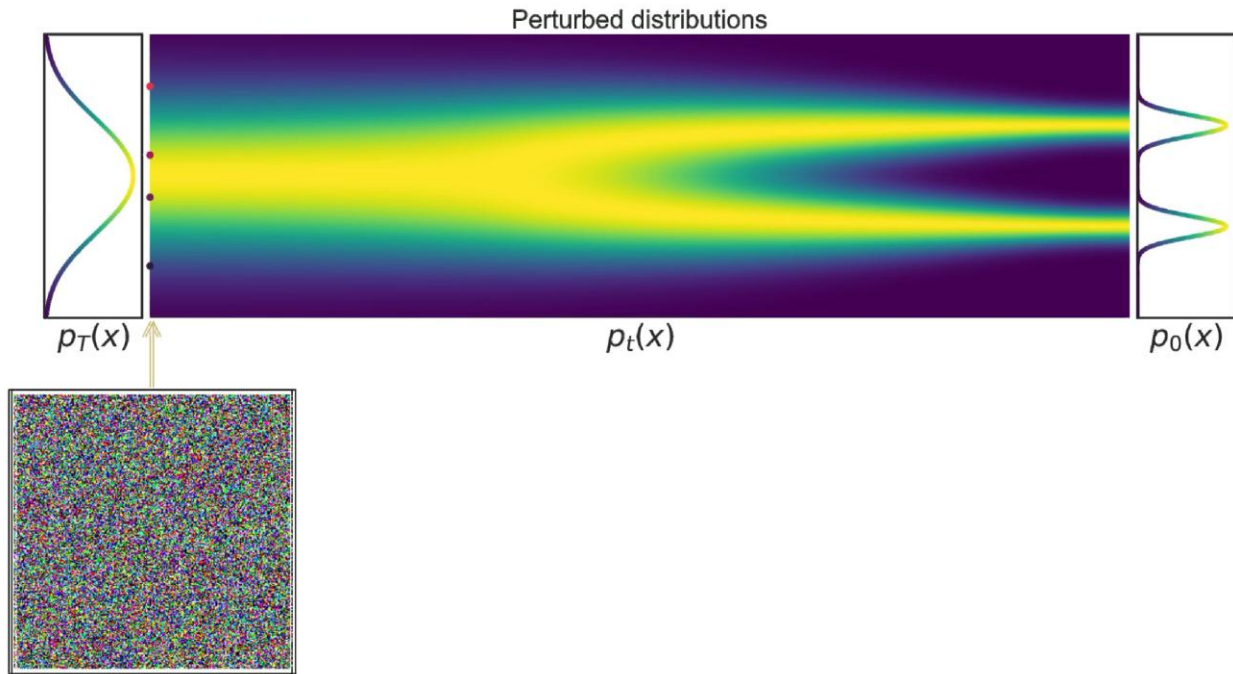
- Minimize objective $\mathbb{E}_{x,t} [\|D_\theta(x_t; \sigma_t) - x\|_2^2]$
or alternatively $\mathbb{E}_{x,\epsilon \sim \mathcal{N}(0,1), \sigma_t} [\|\epsilon_\phi(x_{\sigma_t}; \sigma_t) - \epsilon\|_2^2]$
- This is the training objective of diffusion model!
Note: exact loss formulation can be a bit more complicated, see e.g. Song et al., Appendix E.1

Langevin Dynamics: Forward Diffusion (t: 0 $\rightarrow$ T)

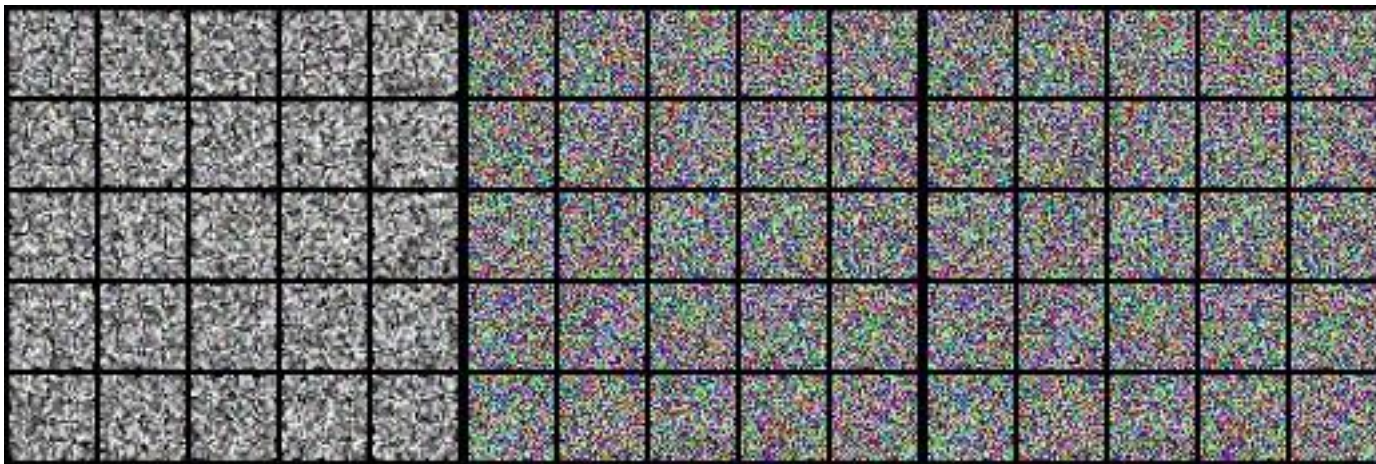
Continuous time SDE formulation



Langevin Dynamics: Reverse Diffusion ($t: T \rightarrow 0$)



Langevin Dynamics: Reverse Diffusion ($t: T \rightarrow 0$)



(Continuous) Diffusion Math via SDE

- Forward diffusion can be described by Ito stochastic differential equation (SDE):

$$d\mathbf{x} = \mathbf{f}(\mathbf{x}, t)dt + g(t)d\mathbf{w}$$

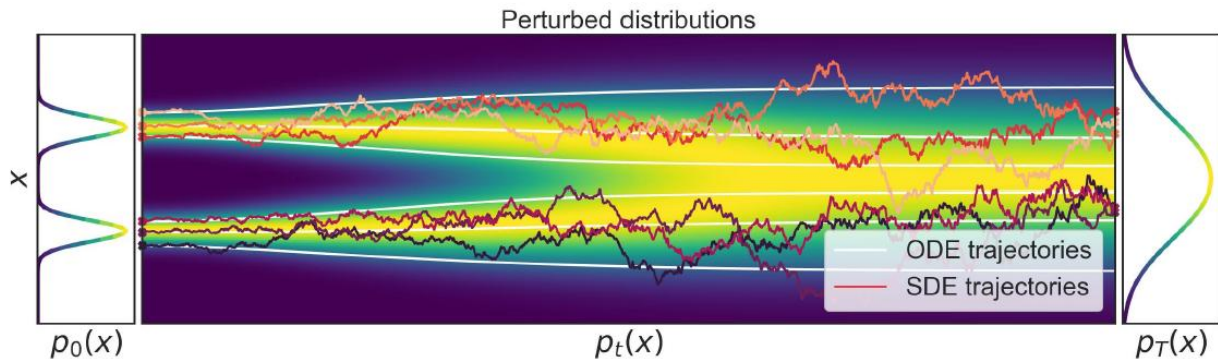
drift coefficient diffusion coefficient noise with variance dt

- Reverse diffusion can be described by Ito SDE:

$$d\mathbf{x} = [\mathbf{f}(\mathbf{x}, t) - g^2(t)\nabla_x \log p_t(\mathbf{x})]dt + g(t)d\tilde{\mathbf{w}}$$

distribution of noisy(!) images at step t standard Wiener process, i.e., noise with variance dt

SDE vs. ODE



SDE
(stochastic differential equation)



Probability Flow ODE
(ordinary differential equation)

$$d\mathbf{x} = -\sigma^2(t)\nabla_x \log p_t(x) dt + \sigma(t)d\tilde{\mathbf{w}}$$

$$d\mathbf{x} = -\sigma^2(t)\nabla_x \log p_t(x) dt$$

Diffusion Model Architectures

Text-to-image Generation Models

OpenAI's DALL-E 1,2,3



Google's Imagen



A chromeplated cat sculpture placed on a Persian rug.



Android Mascot made from bamboo.



Intricate origami of a fox and a unicorn in a snowy forest.

Midjourney



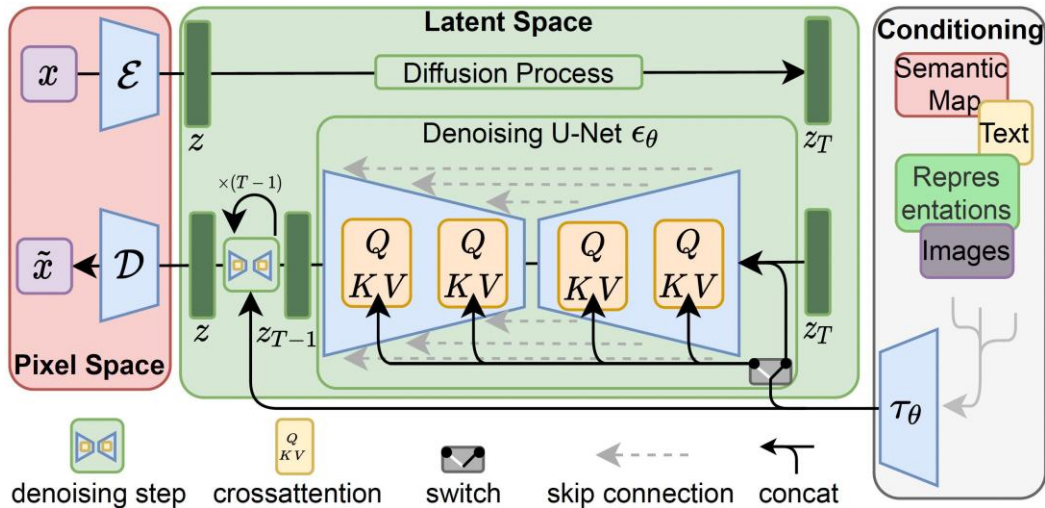
Adobe Firefly



... many more, but many of the models are proprietary ...

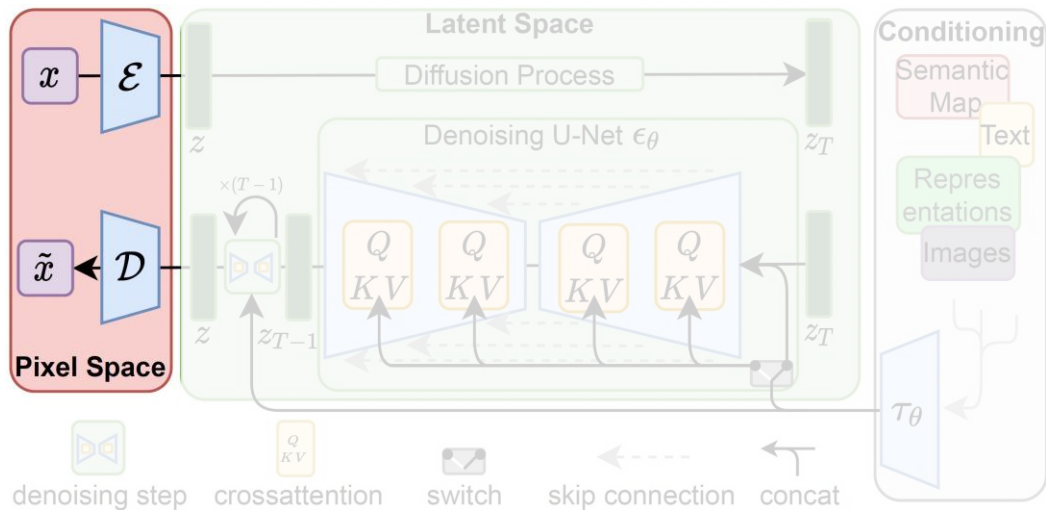
Latent Diffusion Models

1. Variational Autoencoder (VAE) based encoder $\mathcal{E}$ & decoder $\mathcal{D}$
<-> Pixel space is expensive computationally
2. Denoising U-Net ϵ with self attention
3. Conditioning via cross attention



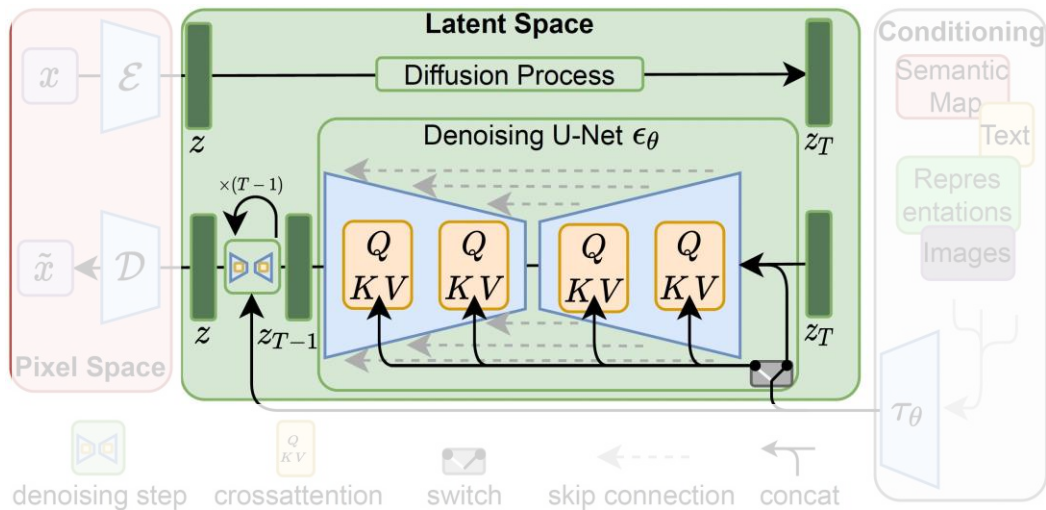
Latent Diffusion Models

1. Variational Autoencoder (VAE) based encoder $\mathcal{E}$ & decoder $\mathcal{D}$
2. Denoising U-Net ϵ_{θ} with self attention
3. Conditioning via cross attention



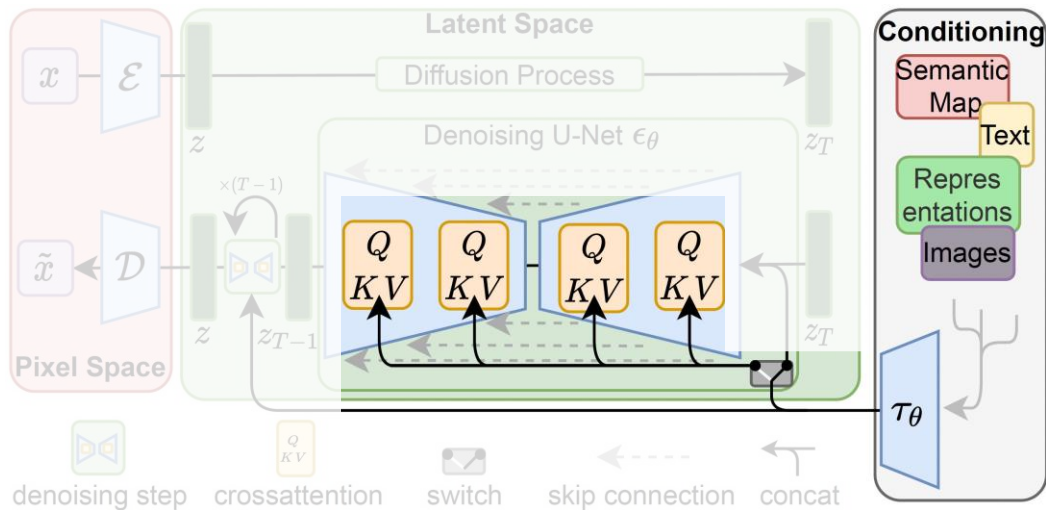
Latent Diffusion Models

1. Variational Autoencoder (VAE) based encoder $\mathcal{E}$ & decoder $\mathcal{D}$
2. Denoising U-Net ϵ_θ with self attention
3. Conditioning via cross attention



Latent Diffusion Models

1. Variational Autoencoder (VAE) based encoder $\mathcal{E}$ & decoder $\mathcal{D}$
2. Denoising U-Net ϵ_θ with self attention
3. Conditioning via cross attention



Reinforcement Learning from Human Feedback



- Align generated images with preferences of humans via reinforcement learning from human feedback / direct preference optimization

References:

DDPM: <https://arxiv.org/pdf/2006.11239>

DDIM: <https://arxiv.org/pdf/2010.02502>

Lab 1 - Diffusion